

Model Selection Report

Date: May 16, 2026

Project: ArenaMCP — opencode.json Agent Brain Assignment

Table of Contents

1. Executive Summary
2. Model Research & Benchmarks
3. Model-to-Agent Assignment Rationale
4. Temperature Configuration Strategy
5. Prompt Adaptations
6. Sources & References

s

1. Executive Summary

This report evaluates four free-tier models available via OpenCode's provider routing and assigns each to the most fitting agent roles in the ArenaMCP project's opencode.json configuration. The assignment is based on benchmark performance, architecture characteristics, context window size, inference speed, and demonstrated strengths in specific task categories (coding, reasoning, agentic workflows, productivity, documentation).

Models evaluated:

- DeepSeek V4 Flash Free (opencode/deepseek-v4-flash-free)
- MiniMax M2.5 Free (opencode/minimax-m2.5-free)
- Nemotron 3 Super Free (opencode/nemotron-3-super-free)
- Qwen3.6 Plus Free (opencode/qwen3.6-plus-free)

2. Model Research & Benchmarks

2.1 DeepSeek V4 Flash Free

Released April 24, 2026 by DeepSeek. A 284B-parameter Mixture-of-Experts model with 13B active parameters per token. Features hybrid Compressed Sparse Attention (CSA) + Heavily Compressed Attention (HCA), a 1M-token context window, and up to 384K output tokens. Licensed under MIT.

Strengths

- Extremely cost-efficient (free tier, \$0.14/M input on paid)
- 1M-token context window — ideal for large codebases and long documents
- Strong coding benchmarks: V4-Pro-Max scores 80.6% SWE-bench Verified, 93.5 LiveCodeBench
- High throughput — 2x speedup over V3 with CSA/HCA attention reducing FLOPs to 27%
- MIT license — fully open weights, commercial use permitted
- Strong reasoning with supported reasoning effort modes (high/xhigh)

Limitations

- Flash variant (13B active) is weaker than Pro (49B active) for complex multi-step reasoning
- Struggles with creativity, nuanced reasoning, and precision tasks per real-world testing
- Text-only — no multimodal (image/audio/video) input support
- Requires cloud API for practical use (4x A100 80GB needed for local inference)
- Knowledge cutoff May 2025 — outdated on very recent events

Key Benchmarks

Benchmark	DeepSeek V4 Flash	Notes
SWE-bench Verified	~78% (est. for Flash)	Pro variant scores 80.6%
LiveCodeBench	93.5 (Pro-Max)	Highest coding score of any model

Source: Morph LLM (morphllm.com/deepseek-v4), OpenRouter (openrouter.ai/deepseek/deepseek-v4-flash:free), Geeky Gadgets (geeky-gadgets.com/open-source-deepseek-v4-limitations), AI Stack Choice (aistackchoice.com/deepseek-v4-review-2026)

2.2 MiniMax M2.5 Free

Released February 12, 2026 by MiniMax. A 230B-parameter MoE model (~10B active). Designed for real-world productivity — trained across 200,000+ real-world digital work environments. 197K-token context window, 8K output tokens. MIT license. Optimized for agentic tasks, office workflows, and tool use.

Strengths

- 99% reliability rate across benchmarks — minimal technical failures
- SWE-bench Verified: 80.2% — competitive with frontier models
- BrowseComp: 76.3% — best-in-class web research capability
- Token-efficient — ~20% fewer reasoning rounds than M2.1
- Strong at: coding (93% accuracy, 82nd percentile), reasoning (94%, 80th percentile)
- Function calling, structured output, and reasoning mode support natively
- Perfect accuracy in General Knowledge and Ethics benchmarks
- Excellent at office/work productivity — Word, Excel, PowerPoint generation

Limitations

- Hallucination benchmark: 88% (35th percentile) — weaker at uncertainty acknowledgment
- Instruction Following: 65% — inconsistent with complex multi-step instructions
- Smaller context window (197K) compared to competitors (1M)
- Text-only — no multimodal input
- Moderate speed — 99 tok/s, 1.17s TTFT
- Newer model with less production track record than DeepSeek

Key Benchmarks

Benchmark	MiniMax M2.5	Notes
SWE-bench Verified	80.2%	Competitive with frontier models
BrowseComp	76.3%	Best-in-class web research
Reliability	99%	Highest in its class

Source: Benchable.ai (benchable.ai/models/minimax/minimax-m2.5-20260211), OpenRouter (openrouter.ai/minimax/minimax-m2.5:free), NVIDIA NGC (catalog.ngc.nvidia.com/orgs/nim/teams/minimax-ai/containers/minimax-m25), LM Market Cap (lmmarketcap.com/model/minimax-m2-5-free)

2.3 Nemotron 3 Super Free

Released March 11, 2026 by NVIDIA at GTC 2026. A 120B-parameter hybrid Mamba-Transformer MoE with only 12B active parameters. Features LatentMoE (512 experts, 22 active), Multi-Token Prediction (MTP) for built-in speculative decoding, and NVFP4 pretraining. Native 1M-token context window. NVIDIA Open Model License.

Strengths

- Fastest inference in class — 2.2x throughput vs GPT-OSS-120B, 7.5x vs Qwen3.5-122B
- 1M-token context with Mamba-2 linear-time complexity — practical long-context
- SWE-bench Verified: 60.47% (OpenHands), RULER 1M: 91.75%, HMMT: 93.67%
- Built for multi-agent systems — handles "context explosion" in agent loops
- Multi-Token Prediction: avg 3.45 token acceptance length (vs R1's 2.70) — 3x wall-clock speedup
- PinchBench (agent brain): 85.6% — top open-weight on agentic benchmarks
- Fully open: weights, training data (10T tokens), recipes, RL environments
- Designed specifically for software dev and cybersecurity multi-agent applications

Limitations

- Lower raw benchmark scores on pure coding (SWE-bench 60.47%) than DeepSeek or Qwen
- NVIDIA Open Model License — not MIT, some restrictions may apply
- Text-only — no multimodal capabilities
- Newer architecture (hybrid Mamba-Transformer) — less ecosystem testing
- 262K context on OpenRouter free tier (vs native 1M)
- Lower active parameters (12B) limits single-step reasoning depth

Key Benchmarks

Benchmark	Nemotron 3 Super	vs GPT-OSS-120B
SWE-bench (OpenHands)	60.47%	41.90%
RULER (1M context)	91.75%	22.30%
PinchBench (agent)	85.6%	Best open-weight

Source: NVIDIA Research (research.nvidia.com/labs/nemotron/Nemotron-3-Super/), NVIDIA Developer Blog (developer.nvidia.com/blog/introducing-nemotron-3-super), StackBuilt AI (stackbuiltai.com/nemotron-3-super-review-2026), LearnAIForge (learnaiforge.com/articles/nvidia-nemotron-3-super-open-model-benchmarks-2026)

2.4 Qwen3.6 Plus Free

Released April 2, 2026 by Alibaba's Qwen team. A closed-weight flagship model combining linear attention with sparse MoE routing. 1M-token context window, up to 66K output tokens. Multimodal (text, image, PDF, video input). Currently available as a free preview on OpenRouter (data collection period).

Strengths

- Terminal-Bench 2.0: 61.6 — beats Claude Opus 4.5 (59.3) for agentic terminal coding
- SWE-bench Verified: 78.8 — approaches Claude's 80.9
- Multimodal input — accepts text, images, PDFs, and video (unique among these 4)
- 1M-token context window and 66K output — massive capacity
- Built-in chain-of-thought reasoning (always-on) — no special prompting needed
- Compatible with Claude Code and Cline toolchains
- Major gains in "vibe coding," front-end development, and repo-level problem solving
- OmniDocBench: 91.2 — excellent document understanding

Limitations

- Preview/closed weights — not fully open source
- Data collection during free preview — do not send confidential data
- Higher TTFT (1.51s) — slower first-token latency
- Chinese ecosystem bias — some tooling may favor Alibaba infrastructure
- Production pricing unknown — free tier may be temporary
- 52.5 tok/s output — slower throughput than Nemotron or DeepSeek Flash
- Requires OpenRouter routing — no direct API without third-party provider

Key Benchmarks

Benchmark	Qwen3.6 Plus	vs Claude 4.5 Opus
Terminal-Bench 2.0	61.6	59.3
SWE-bench Verified	78.8	80.9
OmniDocBench	91.2	87.7

Source: RenovateQR (renovateqr.com/blog/qwen-3-6-plus-review-benchmarks-2026), Apidog (apidog.com/blog/qwen-3-6-free-openrouter), CloudPrice (cloudprice.net/models/alibaba-qwen3-6-plus), VPS SOS (vpssos.com/qwen3.6-plus), Apiyi Blog (help.apiyi.com/en/qwen-3-6-plus-coding-agent-multimodal-benchmark-guide-en)

3. Model-to-Agent Assignment Rationale

Each agent role was analyzed for its primary cognitive demands: reasoning depth, coding ability, long-context requirements, speed requirements, and reliability needs. Models were assigned based on the best fit between these demands and each model's demonstrated strengths.

orchestrator → DeepSeek V4 Flash Free

The orchestrator needs the fastest, most reliable model for routing decisions. DeepSeek V4 Flash offers the best balance of speed, 1M context (for full project awareness), and general-purpose reasoning. Its free tier is most likely to have high uptime. Reference: V4 Flash is recommended as the starting point for "everyday tasks" by AI Stack Choice.

mcp-tutor → DeepSeek V4 Flash Free

Teaching MCP concepts demands patient, clear explanations. DeepSeek V4 Flash's reasoning capabilities and 1M context allow it to reference documentation and provide thorough explanations. Its strong performance on General Knowledge benchmarks supports accurate pedagogical answers.

repo-reader → Qwen3.6 Plus Free

Reading and analyzing entire code repositories demands maximum context capacity and strong code understanding. Qwen3.6 Plus offers 1M context and the best Terminal-Bench score (61.6), indicating superior code analysis capabilities. Its multimodal input also helps with reading screenshots of code or architecture diagrams.

architecture-lead → DeepSeek V4 Flash Free

System architecture design requires strong reasoning, 1M context for architecture docs, and balanced judgment. DeepSeek V4 Flash's general-purpose strength and MIT license make it ideal for this role. Its 93.5 LiveCodeBench score demonstrates top-tier reasoning.

build-navigator → MiniMax M2.5 Free

Breaking projects into phased steps requires structured thinking and real-world productivity awareness. MiniMax M2.5 was explicitly designed for "real-world productivity" and "office-work style workflows." Its 99% reliability ensures consistent planning output. Reference: NVIDIA NGC describes M2.5 as optimized for "agentic tasks including software engineering, tool use, search, and office-work style workflows."

backend-engineer → Qwen3.6 Plus Free

Implementing HTTP clients and API integration demands top-tier coding ability. Qwen3.6 Plus is the strongest coding agent among these four, beating Claude Opus 4.5 on Terminal-Bench 2.0. Its 1M context handles full API documentation in-context. Compatibility with Claude Code toolchains ensures smooth integration.

tool-designer → MiniMax M2.5 Free

Designing MCP tool schemas requires structured outputs, reliability, and consistent JSON/function-calling behavior. MiniMax M2.5 has native function calling, structured output support, and 99% reliability — ideal for precise schema design work.

code-reviewer → Nemotron 3 Super Free

Code review demands fast, methodical analysis across multiple dimensions. Nemotron 3 Super's built-in speculative decoding (3.45 avg acceptance length) provides the fastest review cycles. Its multi-agent architecture makes it excellent at catching cross-cutting issues. Strong at PinchBench (85.6%) for agentic reasoning tasks.

test-engineer → Nemotron 3 Super Free

Writing and running tests requires methodical iteration and fast feedback loops. Nemotron 3 Super's 2.2x throughput advantage means faster test generation and validation. Its agentic benchmark leadership suggests strong multi-step test scenario planning.

documentation-writer → MiniMax M2.5 Free

Documentation requires clear communication and real-world productivity focus. MiniMax M2.5 excels at "office-work style workflows" and was trained on real-world digital environments. Its 94% reasoning score supports accurate technical writing. The model's focus on "generating Word, Excel, and PowerPoint" aligns perfectly with documentation tasks.

product-strategist → MiniMax M2.5 Free

Strategic product decisions benefit from structured thinking, reliability, and productivity awareness. MiniMax M2.5's 99% reliability and strength in "complex agentic tasks including software engineering, tool use, search" make it ideal for prioritization and roadmap planning.

architecture-sentinel → Nemotron 3 Super Free

Consistency checking requires fast, repeated analysis across the codebase. Nemotron 3 Super's speed advantage and strong agentic benchmarks make it effective at detecting design drift. Its 1M context supports holistic architecture awareness. Built for "multi-agent systems" where coherence across agents is critical.

Assignment Summary

Agent	Model	Rationale Key
orchestrator	DeepSeek V4 Flash Free	Speed + reliability + 1M context
mcp-tutor	DeepSeek V4 Flash Free	General reasoning + pedagogy
repo-reader	Qwen3.6 Plus Free	Best code analysis + 1M context
architecture-lead	DeepSeek V4 Flash Free	System reasoning + breadth
build-navigator	MiniMax M2.5 Free	Structured planning + reliability

backend-engineer	Qwen3.6 Plus Free	Best coding agent + tooling compat
tool-designer	MiniMax M2.5 Free	Structured output + function calling
code-reviewer	Nemotron 3 Super Free	Speed + agentic multi-step check
test-engineer	Nemotron 3 Super Free	Fast iteration + methodical
documentation-writer	MiniMax M2.5 Free	Office/document workflow specialist
product-strategist	MiniMax M2.5 Free	Productivity + 99% reliability
architecture-sentinel	Nemotron 3 Super Free	Fast consistency + agent coherence

4. Temperature Configuration Strategy

Temperature settings are chosen based on each agent's task profile. Lower temperatures (0.0-0.2) favor deterministic, precise outputs ideal for coding and analysis. Medium temperatures (0.3-0.4) balance creativity and precision for planning and design. Higher temperatures (0.5-0.7) support creative and explanatory writing.

Agent	Temperature	Rationale
orchestrator	0.3	Low, not zero — allows slight routing flexibility while staying decisive
mcp-tutor	0.5	Medium — enables clear, natural explanations without being overly creative
repo-reader	0.2	Low — accuracy and precision in code analysis is critical
architecture-lead	0.3	Low-medium — design needs some creative flexibility with structure
build-navigator	0.4	Medium-low — planning benefits from structured creativity
backend-engineer	0.1	Very low — precise code generation, avoid hallucinations
tool-designer	0.3	Low — schema design needs precision with some flexibility
code-reviewer	0.1	Very low — deterministic analysis, flag issues consistently
test-engineer	0.1	Very low — methodical test

		generation, minimize variance
documentation-writer	0.5	Medium — natural language needs some creative range
product-strategist	0.4	Medium-low — strategic thinking with structured output
architecture-sentinel	0.2	Low — consistent detection of design drift

5. Prompt Adaptations

For each agent, the base prompt was kept intact but a brief model-specific adaptation note was appended to leverage the assigned model's strengths. The key principle: do not rewrite working prompts — instead, add a contextual hint that helps the model perform optimally in its assigned role.

orchestrator (DeepSeek V4 Flash Free)

Appended: "You have excellent routing judgment with broad general knowledge. Use your 1M context awareness to keep the full project state in view."

mcp-tutor (DeepSeek V4 Flash Free)

Appended: "You have strong explanatory reasoning. Use your general knowledge breadth to provide accurate, beginner-friendly MCP explanations."

repo-reader (Qwen3.6 Plus Free)

Appended: "You excel at code analysis and can process full repository context. Leverage your 1M-token capacity to read all files comprehensively."

architecture-lead (DeepSeek V4 Flash Free)

Appended: "You have strong system-level reasoning. Use your broad knowledge to design clean, modular architectures."

build-navigator (MiniMax M2.5 Free)

Appended: "You are optimized for real-world productivity workflows. Break plans into clear, structured, actionable phases."

backend-engineer (Qwen3.6 Plus Free)

Appended: "You excel at agentic coding tasks including API client implementation. Write production-quality, robust code."

tool-designer (MiniMax M2.5 Free)

Appended: "You have strong structured output and function calling capabilities. Define precise, well-typed tool schemas."

code-reviewer (Nemotron 3 Super Free)

Appended: "You perform fast, multi-step analysis across dimensions. Leverage your speed to check all aspects thoroughly."

test-engineer (Nemotron 3 Super Free)

Appended: "You excel at methodical agentic workflows. Generate comprehensive tests with fast iteration."

documentation-writer (MiniMax M2.5 Free)

Appended: "You specialize in real-world productivity and clear document generation. Create practical, beginner-friendly documentation."

product-strategist (MiniMax M2.5 Free)

Appended: "You have strong structured planning capabilities. Prioritize features methodically based on user value and effort."

architecture-sentinel (Nemotron 3 Super Free)

Appended: "You are built for multi-agent coherence checking. Detect design drift quickly and consistently."

6. Sources & References

DeepSeek V4 Flash Free

- Morph LLM — "DeepSeek V4: Architecture, Benchmarks, and API Guide" (April 24, 2026) — morphllm.com/deepseek-v4
- OpenRouter — "DeepSeek V4 Flash (free)" — openrouter.ai/deepseek/deepseek-v4-flash:free
- Geeky Gadgets — "Deepseek V4: Why Its 1.6 Trillion Parameters Aren't Quite Enough" (April 25, 2026) — geeky-gadgets.com/open-source-deepseek-v4-limitations
- AI Stack Choice — "DeepSeek V4 Review 2026: Benchmarks, Pricing, Architecture" (April 30, 2026) — aistackchoice.com/deepseek-v4-review-2026-benchmarks-pricing-architecture
- DeepSeek V4 Network — deepseekv4.network/models
- Hugging Face — huggingface.co/Intel/DeepSeek-V4-Flash-W4A16-AutoRound

MiniMax M2.5 Free

- Benchable.ai — "MiniMax M2.5 Model Details & Benchmarks" — benchable.ai/models/minimax/minimax-m2.5-20260211
- OpenRouter — "MiniMax M2.5 (free) Performance Metrics" — openrouter.ai/minimax/minimax-m2.5:free
- NVIDIA NGC — "MiniMax-M2.5 NIM Container" — catalog.ngc.nvidia.com/orgs/nim/teams/minimax-ai/containers/minimax-m25
- LM Market Cap — "MiniMax M2.5 (free) — Pricing & Benchmarks 2026" — lmmarketcap.com/model/minimax-m2-5-free
- LLM Stats — "MiniMax M2.5 Benchmarks" — llm-stats.com/models/minimax-m2.5
- Galaxy.ai — "MiniMax M2.5 Model Specs" — galaxy.ai/model/minimax-m2-5
- MiniMax-M2 Official Site — minimax-m2.com/minimax-m25

Nemotron 3 Super Free

- NVIDIA Research — "NVIDIA Nemotron 3 Super" (March 10, 2026) — research.nvidia.com/labs/nemotron/Nemotron-3-Super/
- NVIDIA Developer Blog — "Introducing Nemotron 3 Super" (March 12, 2026) — developer.nvidia.com/blog/introducing-nemotron-3-super

- StackBuilt AI — "NVIDIA Nemotron 3 Super Review 2026" (March 21, 2026) — stackbuiltai.com/nemotron-3-super-review-2026
- LearnAIForge — "Nemotron 3 Super: Open Model Agentic Benchmarks" (March 23, 2026) — learnaiforge.com/articles/nvidia-nemotron-3-super-open-model-benchmarks-2026
- Artificial Analysis — "NVIDIA Nemotron 3 Super 120B A12B" — artificialanalysis.ai/models/nvidia-nemotron-3-super-120b-a12b
- OpenRouter — "Nemotron 3 Super (free)" — openrouter.ai/nvidia/nemotron-3-super-120b-a12b:free
- LLM Stats — "Nemotron 3 Super Technical Deep Dive" (March 11, 2026) — llm-stats.com/blog/research/nemotron-3-super-launch
- Hugging Face — huggingface.co/nvidia/NVIDIA-Nemotron-3-Super-120B-A12B-FP8

Qwen3.6 Plus Free

- RenovateQR — "Qwen 3.6 Plus Review: Benchmarks, Architecture" (April 2, 2026) — renovateqr.com/blog/qwen-3-6-plus-review-benchmarks-2026
- Apidog — "Qwen 3.6 Available on OpenRouter" (March 31, 2026) — apidog.com/blog/qwen-3-6-free-openrouter
- CloudPrice — "Qwen3.6 Plus pricing & specs" — cloudprice.net/models/alibaba-qwen3-6-plus
- VPS SOS — "Qwen3.6 Plus Preview Is Free on OpenRouter" (March 31, 2026) — vpssos.com/qwen3.6-plus
- Apiyi Blog — "Qwen3.6-Plus In-depth Interpretation" (April 3, 2026) — help.apiyi.com/en/qwen-3-6-plus-coding-agent-multimodal-benchmark-guide-en
- OpenRouter — "Qwen3.6 Plus" — openrouter.ai/qwen/qwen3.6-plus
- OpenRouter — "Qwen3.6 Plus (free) Performance" — openrouter.ai/qwen/qwen3.6-plus/performance
- YingTu AI — "Which Qwen Model Should You Use?" (May 8, 2026) — yingtu.ai/en/blog/qwen-model-guide-2026

General

- OpenCode Model Registry — deepwiki.com/opencode-ai/opencode/3.1-model-registry
- AkitaOnRails — "LLM Coding Benchmark May 2026" (April 24, 2026) — akitaonrails.com/en/2026/04/24/llm-benchmarks-parte-3-deepseek-kimi-mimo
- Galaxy.ai — Model Comparisons — galaxy.ai/blog/compare
- Artificial Analysis — Model Comparisons — artificialanalysis.ai